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Insulin: Insulin is a peptide hormone and plays an important role in glucose homeostasis. It's main effect is to lower blood glucose levels by increasing the uptake of glucose into the body cells, especially muscle and fat cells. Insulin also inhibits the breakdown of glycogen to glucose, the conversion of amino acids or fats to glucose, so insulin is rightly called a hypoglycemic hormone.

Humulin N: Human insulin is produced by recombinant DNA technology (genetic engineering) and administered to diabetic patients as injection and not by oral consumption. Reason: Digestive enzymes digest it.

Glucagon: Glucagon is a polypeptide hormone. It is a potent hyperglycaemic hormone that acts on the liver and promotes the breakdown of glycogen to glucose (Glygogenolysis), synthesis of glucose from lactic acid and from non-carbohydrate molecules (Gluconeogenesis). Glucagon releases glucose from the liver cells, increasing the blood glucose levels. Since glucagon reduces the cellular uptake and utilisation of glucose it is called a hyperglycemic hormone. Prolonged hyperglycemia leads to the disorder called diabetes mellitus.

Insulin: The half life period of insulin (in plasma) is 6 minutes

It is cleared from the circulation within 10-15 minutes

Endocrine glands control and coordinate the body functions through secreting certain chemical messengers called hormones. Due to certain physiological reasons, the blood glucose level of an otherwise normal person.

a. Give the possible cause for the increases in blood glucose level.

b. What is the chemical nature of this hormone? Discuss its role in the body.

c. How can this condition be reversed?

11.2.9 Gonads

Testis: A pair of testis is present in the scrotal sac of males. The testis functions as a sex organ and also as an endocrine gland. The testis is composed of seminiferous tubules and interstitial cells or Leydig cells. The Leydig cells secrete several male sex hormones, collectively called androgens, mainly testosterone.

Functions of testosterone: Under the influence of FSH and LH, testosterone initiates maturation of male reproductive organs, and the appearance of secondary sexual characters, muscular growth, growth of facial and axillary hair, masculine voice and male sexual behaviour. It enhances the total bone matrix and plays a stimulating role in the process of spermatogenesis.

Ovary: Females have a pair of ovaries located in the pelvic region of